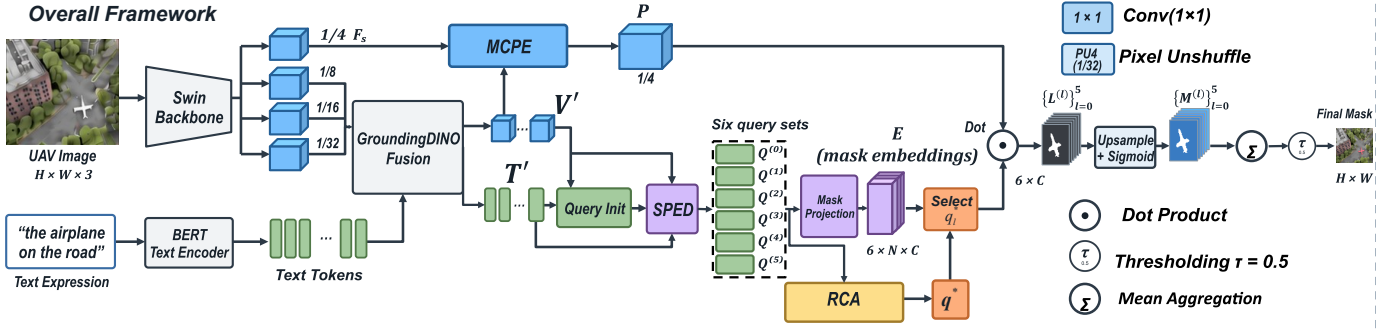

Overall Framework
UAV Image
H × W × 3
Swin Backbone
1/4 F_s
1/8
1/16
1/32
GroundingDINO Fusion
MCPE
P
1/4
V'
T'
"the airplane on the road"
Text Expression
BERT Text Encoder
Text Tokens
Query Init
SPED
Six query sets
$Q^{(0)}$
$Q^{(1)}$
$Q^{(2)}$
$Q^{(3)}$
$Q^{(4)}$
$Q^{(5)}$
Mask Projection
E
(mask embeddings)
6 × N × C
Select q_i
RCA
q^*
Dot
6 × C
$\{L^{(l)}\}_{l=0}^{5}$
Upsample + Sigmoid
$\{M^{(l)}\}_{l=0}^{5}$
Final Mask
H × W
1 × 1 Conv(1×1)
PU4 (1/32) Pixel Unshuffle
Dot Product
Thresholding τ = 0.5
Mean Aggregation